

# Trevor W. Exley

Via Morego, 30, 16163 Genova, Italy

trevor.exley@iit.it | exley.info

Google Scholar | ORCID: 0000-0003-3175-0587



## Personal Statement

Soft robotics researcher specializing in architected, monolithic systems where structure and function are co-designed. My current work investigates material architectures that integrate actuation and sensing within a single body, enabling pathways toward printable, assembly-free soft robotic systems.

## Education

**Ph.D., Biomedical Engineering**, University of North Texas 2021–2024  
Dissertation: “Thermo-Reversible Phase-Change Actuators for Physical Human–Robot Interactions. Supervisor: Amir Jafari. ” GPA: 4.00/4.00.

**M.S., Biomedical Engineering**, University of North Texas 2020–2021  
Thesis: “Parkinson’s Disease and UPDRS-III Prediction Using Quiet Standing Data and Applied Machine Learning.”

**B.S., Biomedical Engineering**, University of North Texas 2017–2020  
Minors: Mathematics, Mechanical Engineering.  
Senior Capstone: “Hercules: Low Cost Trans-Humeral Prosthetic Capable of Power and Precision Grip Controlled Through Foot and Upper-Body Movement”

## Appointments

**Postdoctoral Researcher**, Soft Biorobotics Perception Research Line, Istituto Italiano di Tecnologia (IIT), Genova 2024–Present  
*Monolithic soft robotics; integrated actuation, sensing & fabrication.*  
Supervisor: Lucia Beccai

**PhD Researcher**, Advanced Robotic Manipulators Lab, University of North Texas 2021–2024  
*Thermo-active variable impedance actuators; soft robotic systems.*  
Supervisor: Amir Jafari

**Lab Affiliated**, Biomedical AI Lab, University of North Texas 2020–2023  
*Biomedical machine learning; wearable sensor data.*  
Supervisor: Mark V. Albert

## Peer-Reviewed Publications

(\* = corresponding author; † = co-corresponding author)

## Submitted / Under Revision

- [C.8] **T. Exley**, A. Coutinho, L. Beccai. Selective Unit-Cell Actuation in Lattice Structures for Distributed Morphology in Soft Robots. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2026) (accepted)
- [C.7] P. Trunin, C. Gay, A. B. Nardin, **T. Exley**, D. Cafiso, L. Beccai. Bidirectional Optical Sensors for Actuation Tracking (BOAT) in Soft Lattice Systems. *2026 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (accepted)
- [B.1] D. Cafiso, **T. Exley**, P. Trunin, A. B. Nardin, S. Hajaresab Ladakhana, L. Beccai. Monolithic Strategies for Soft Robotics via Light-Induced 3D printing: Towards Fully Integrated Systems. Chapter in *Digital Light Processing and Stereolithography Additive Manufacturing*. (submitted)

#### In Press / Published

- [J.13] D. Johnson, **T. Exley**, M.-A. Torres, J. Slayton, M. Ecker, A. Jafari. Tailoring Silicone Mixtures for Soft Robotics: Predictive Modeling and Experimental Validation in Pneumatic Soft Actuators. *Bioinspiration & Biomimetics* (2026)
- [C.6] **T. Exley**, A. B. Nardin, P. Trunin, D. Cafiso, L. Beccai. Monolithic Units: Actuation, Sensing, and Simulation for Integrated Soft Robot Design. *IEEE RoboSoft* (2026)
- [J.12] P. Trunin, D. Cafiso, A. B. Nardin, **T. Exley<sup>†</sup>**, L. Beccai. MELEGROS: Monolithic Elephant-Inspired Gripper with Optical Sensors. *Advanced Science* (2026). **[Featured on Back Cover]**
- [J.11] **T. Exley**, D. Johnson, A. Jafari. Estimating Stiffness and Damping of a Novel Variable Impedance Actuator based on Adjusting Viscoelastic Properties of Thermoresponsive Polycaprolactone in Harmonic Motions. *Scientific Reports*. (2025)
- [J.10] **T. Exley**, R. Wijesundara, S. Wang, A. Moridani, T. Nilforooshan, A. Jafari. TVIM: Thermoactive Variable Impedance Module Evaluating Shear-Mode Capabilities of Polycaprolactone. *IEEE Access* (2025).
- [C.5] J. Jenkins, O. Madera, C. Guerrero, K. Humes, A. Malmquist, C. Renfrew, Q. Bakker, K. Siu, **T. Exley**, A. Jafari. Design and Implementation of a High-Precision Motor Control System for Adjustable-Stiffness Biomedical Treadmills. In *Proc. IEEE 18th Dallas Circuits and Systems Conf. (DCAS)*, 1–6 (2025).
- [C.4] F. Tajomi, **T. Exley**, A. Jafari. Development of a Miniature Thermal-Based Variable Impedance Actuator Using Thermoplastic Polymers for Scalable and Compact Robotic Applications. In *Proc. IEEE 18th Dallas Circuits and Systems Conf. (DCAS)*, 1–6 (2025).
- [C.3] **T. Exley**, R. Wijesundara, N. Tan, A. Sunkara, X. He, S. Wang, B. Chan, A. Jain, L. Espinosa, A. Jafari. Agonist–Antagonist Pouch Motors: Bidirectional Soft Actuators Enhanced by Thermally Responsive Peltier Elements. In *IEEE/RSJ IROS* (2024).
- [C.2] K. Ruiz, S. Ryan, S. Stutsman, T. Tirumala, J. Williams, R. Wijesundara, **T. Exley**, A. Jafari. Design considerations of Peltier-Integrated Therapeutic Wrist Wrap for Medical Applications. *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (2024)
- [J.9] **T. Exley**, D. Johnson, A. Jafari. Comparative Analysis of Peltier Devices and Flexible Heater Strips for Enhancing Bandwidth in Thermo-Active Soft Actuators. *Journal of Medical Robotics Research* 9(03n04):2450002 (2024).

- [J.8] **T. Exley**, E. Hays, D. Johnson, A. Moridani, R. Motati, A. Jafari. Toward a Unified Naming Scheme for Thermo-Active Soft Actuators: A Review of Materials, Working Principles, and Applications. *Robotics Reports* 2(1):15–28 (2024).
- [J.7] **T. Exley**, D. Johnson, A. Jafari. A Novel Variable Impedance Actuator Utilizing Adjustable Viscoelastic Properties of Thermoresponsive Polycaprolactone. *Robotics Reports* 1(1):57–66 (2023).
- [J.6] **T. Exley**, D. Johnson, A. Jafari. Towards a Novel Thermal-Based Variable Impedance Module through Adjusting Viscoelastic Properties of a Thermoresponsive Polymer. *IEEE Transactions on Medical Robotics and Bionics*, 1–1 (2023).
- [J.5] **T. Exley**, D. Johnson, A. Jafari. Utilizing the Peltier Effect for Actuation of Thermo-Active Soft Robots. *Smart Materials and Structures* (2023).
- [J.4] E. Hays, J. Slayton, G. Tejeda-Godinez, E. Carney, K. Cruz, **T. Exley**, A. Jafari. A Review of Rehabilitative and Assistive Technologies for Upper-Body Exoskeletal Devices. *Actuators* 12 (Soft Robotics in Biomedical Application):178 (2023).
- [C.1] Z. Liu, **T. Exley**, A. Meek, R. Yang, H. Zhao, M. V. Albert. Predicting GPU Performance and System Parameter Configuration Using Machine Learning. In *IEEE ISVLSI*, 253–258 (2022).
- [J.3] **T. Exley**, S. Moudy, R. M. Patterson, J. Kim, M. V. Albert. Predicting UPDRS Motor Symptoms in Individuals with Parkinson’s Disease from Force Plates Using Machine Learning. *IEEE Journal of Biomedical and Health Informatics* 26(7):3486–3494 (2022).
- [J.2] **T. Exley**, A. Jafari. Increasing Robustness and Output Performance of Variable Stiffness Actuators in Periodic Motions. *Mechanism and Machine Theory* 169:104645 (2022).
- [J.1] **T. Exley**, A. Jafari. Maximizing Energy Efficiency of Variable Stiffness Actuators through an Interval-Based Optimization Framework. *Sensors and Actuators A: Physical* 332:113123 (2021).

## Grants & Funding

NIH G-RISE (T32GM136501)

2021-2024

Amount: \$109,318 USD.

## Patents

- [P.4] A. Jafari, **T. W. Exley**. **Thermoactivate modular soft actuator based on phase transition**. US Patent No. 12,384,024, issued Aug. 12, 2025.
- [P.3] R. Kaleigh, S. Ryan, S. Stutsman, T. Tirumala, J. Williams, R. Wijesundara, **T. Exley**, A. Jafari. **Peltier-integrated therapeutic wrap**. US Patent App. 18/649,757 (2024).
- [P.2] E. Hays, E. Carney, K. Cruz, J. Slayton, G. Tejeda-Godinez, **T. Exley**, A. Jafari. **Devices for biomechanical assistance including an exoskeleton and a robotic glove**. US Patent App. 18/649,708 (2024).
- [P.1] A. Jafari, **T. W. Exley**. **Thermal-based variable impedance actuator**. US Patent App. 18/418,008 (2024).

## Awards

- [4] Outstanding Ph.D. Graduate, University of North Texas (2024)
- [3] Outstanding M.S. Student, University of North Texas (2021)
- [2] Outstanding Senior; Distinguished Honors College Scholar; UNT Excellence Scholarship (2020)
- [1] Esports Scholarship, UNT Varsity League of Legends Team (2019–2020)

## Projects

### Deliverables, presentations, multi-partner coordination

#### **PROBOSCIS (PROBOSCIdean Sensitive Soft Robot for Versatile Gripping)** 2024–2025

*European Union Horizon Grant, Agreement No. 863212.*

## Teaching

### Traditional Courses

- [1] Teaching Fellow, UNT BMEN 2320 – *Biomedical Instrumentation* Summer 2021
- [2] Teaching Assistant, UNT BMEN 2210 – *Biomedical Data Acquisition*; BMEN 2320 – *Biomedical Instrumentation* 2020–2021

### External courses, tutorials, and workshops

- [3] **Soft Robotics in i4.0**, Robotics in Industry 4.0: Challenges and Opportunities  
*University of Texas at San Antonio, Summer 2022*

## Research Supervised

Mentored >14 students individually as primary supervisor (now at **Harvard University, UC Berkeley, UT Austin, Georgia Tech, Rice University, Texas A&M**, University of North Texas, University of St. Thomas), plus 3 senior capstone groups at University of North Texas. Mentored students co-authored peer-reviewed conference papers [C.3], [C.5] and journal articles [J.8], and contributed to patent applications.

## Service

### Reviewer

Actuators; Applied Sciences; Biomimetics; IEEE Access; IEEE BioRob; IEEE ICORR; IEEE/RSJ IROS; IEEE RoboSoft; IEEE RA-L; IEEE T-MRB; IEEE T-RO; JNER; Robotics Reports; Soft Robotics

## Media Coverage

- [2] H. Noel. *UNT Soft Robotics Lab Developing Patent-Pending Inventions*. University of North Texas Research & Innovation News, Apr 25, 2024. (*Featured subject; profiles two patents.*)

- [1] J. DeLeón. *Engineering Better Health*. The North Texan (UNT alumni magazine), Fall 2021. (*Profiled.*)

## Professional Memberships

|                                                          |              |
|----------------------------------------------------------|--------------|
| Institute of Electrical and Electronics Engineers (IEEE) | 2020–Present |
| Biomedical Engineering Society (BMES)                    | 2020–2024    |
| American Society of Biomechanics (ASB)                   | 2020-2021    |

## References

Lucia Beccai, Ph.D., Senior Researcher, Istituto Italiano di Tecnologia. [lucia.beccai@iit.it](mailto:lucia.beccai@iit.it)  
Amir Jafari, Ph.D., Associate Professor, University of North Texas. [amir.jafari@unt.edu](mailto:amir.jafari@unt.edu)  
Vijay Vaidyanathan, Ph.D., Founding Chair of Biomedical Engineering, University of North Texas. [Vijay.Vaidyanathan@unt.edu](mailto:Vijay.Vaidyanathan@unt.edu)